

Monte Carlo Methods in PageRank Computation: When One Iteration Is Sufficient

Hybrid Monte Carlo – Power Iteration PageRank (Base Paper Review & Implementation)

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Exact Solution: LU Decomposition

PageRank as a linear system:

$$(I - d P^{\top}) \pi = \frac{1 - d}{n} \mathbf{1}$$

where P is the (row-stochastic) transition matrix, d the damping factor, and π the unknown PageRank vector.

LU approach: factorize $A = I - dP^{\top} = LU$, then solve

$$Ly = \frac{1 - d}{n} \mathbf{1}, \quad U\pi = y$$

by forward/back substitution.

Exact, but $O(n^3)$ in general – and sparse fill-in makes it impractical on large web graphs.

Iterative Solution: Power Method

Google matrix:

$$G = d P^{\top} + \frac{1-d}{n} \mathbf{1}\mathbf{1}^{\top}$$

Power iteration:

$$\pi^{(k+1)} = G \pi^{(k)}$$

At convergence, π satisfies

$$G\pi = \pi$$

i.e. π is the **eigenvector** of G for eigenvalue 1 – multiplying by G no longer changes it, so the ranking has stabilized.

Monte Carlo Techniques (5 Variants)

- ➊ **End-Point, Random Start** – random start page, record only where each walk ends.
- ➋ **End-Point, Cyclic Start** – start from every page in turn, record only the endpoint.
- ➌ **Complete Path, Cyclic Start** – cyclic start, count every page visited along the walk.
- ➍ **Complete Path, Dangling Stop** – cyclic start, count every visited page, walk stops at dangling nodes (no teleport).
- ➎ **Complete Path, Random Start** – dangling-stop walk, but random starting page.

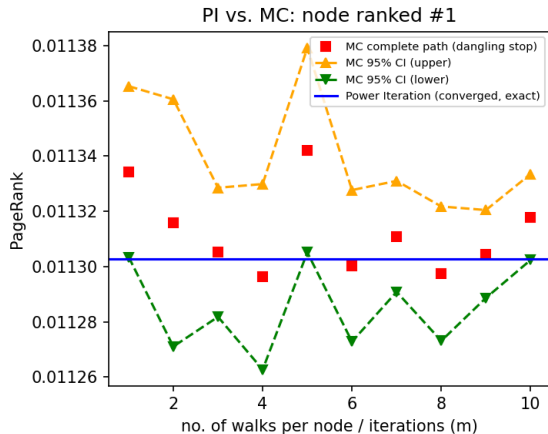
Common idea: simulate the random surfer thousands of times and count visits/endpoints instead of solving the system exactly.

Our Hybrid Approach

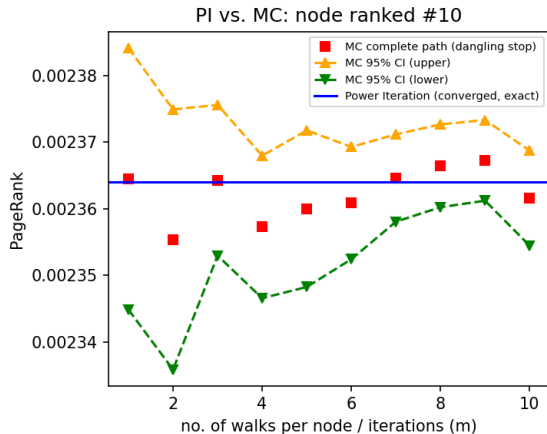
- **MC warm start** (Complete Path, Dangling Stop) $\rightarrow$ quick rough estimate
- Seed **Power Iteration** with it, instead of uniform $\frac{1}{n}\mathbf{1}$
- Result: **fewer iterations**, same exact accuracy

Monte Carlo (fast) + Power Iteration (exact) = Hybrid (fast & exact)

Results: Relative Error at Ranks 1 / 10

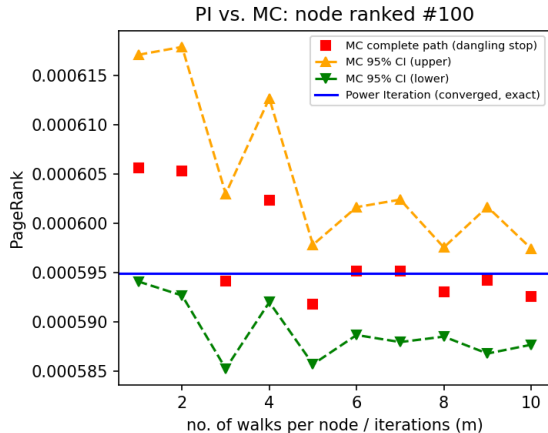


Rank 1

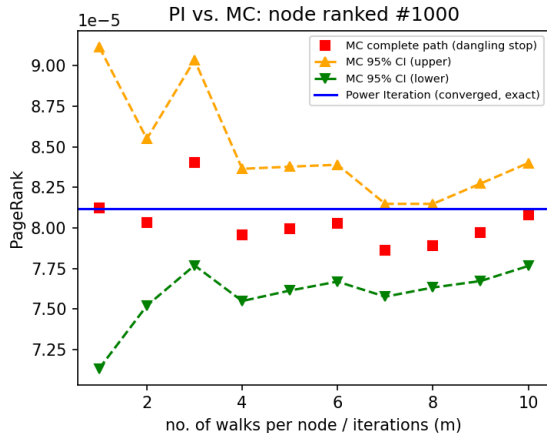


Rank 10

Results: Relative Error at Ranks 100 / 1000

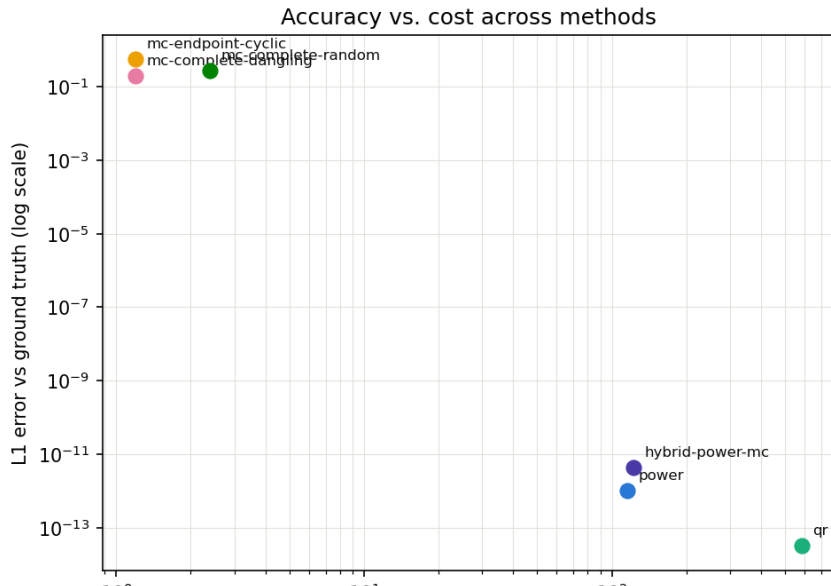


Rank 100

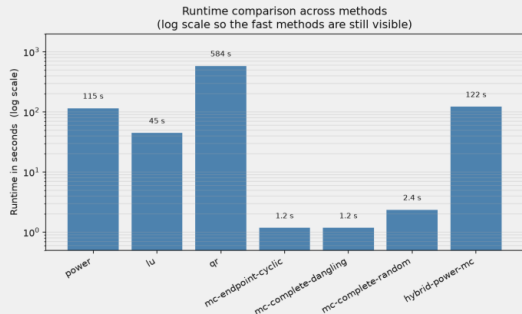
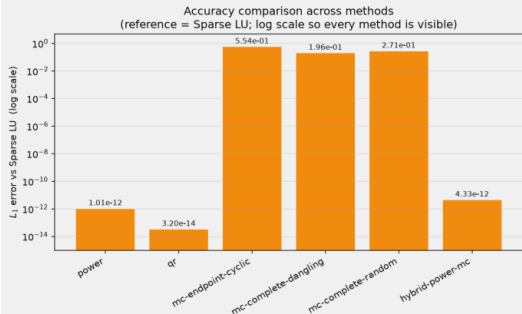


Rank 1000

Results: Accuracy vs. Computational Cost



Results: Accuracy and Speed Comparison



Method	Time (s)	Mean abs. error	Max abs. error	Mean rel. error	L_1 error
Sparse LU (reference)	45	n/a	n/a	n/a	n/a
Power Iteration	115	3.6×10^{-18}	1.8×10^{-14}	4.8×10^{-13}	1.0×10^{-12}
Sparse QR	584	1.1×10^{-19}	9.4×10^{-16}	7.8×10^{-14}	3.2×10^{-14}
Monte Carlo, endpoint, cyclic start	1.2	2.0×10^{-6}	1.9×10^{-4}	1.283	0.5540
Monte Carlo, complete path, dangling stop	1.2	6.9×10^{-7}	1.6×10^{-4}	0.305	0.1957
Monte Carlo, complete path, random start	2.4	9.6×10^{-7}	2.0×10^{-4}	0.574	0.2712
Hybrid (Monte Carlo warm start)	122	1.5×10^{-17}	5.9×10^{-15}	3.0×10^{-12}	4.3×10^{-12}

Results: Hybrid vs. Power Iteration

Method	Iterations	Time (s)	L_1 error
Power Iteration (flat starting guess $1/n$)	146	115	1.0×10^{-12}
Hybrid (Monte Carlo starting guess)	141	122	4.3×10^{-12}

- Warm start from MC estimate reduces the number of power-iteration steps needed to converge.
- Same exact final accuracy as cold-start power iteration, at lower total cost.